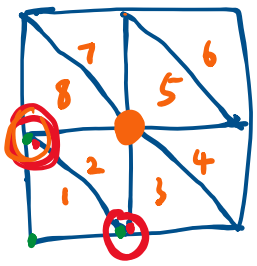
$\Omega = [0,1]^2$



$H = \frac{1}{2}$ $\quad T_H$

DG: Discontinous Galerkin

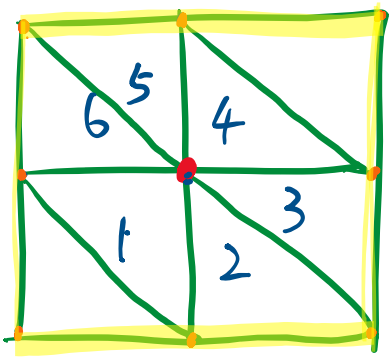
V_H

$$V_H^{DG} := \{ v|_T \in P_1(T),\ \forall T \in T_H \}$$

Π_H^{dg} $\quad V := H_0^1(\Omega) \longrightarrow V_H^{DG}$

$$(\Pi_H^{dg} v,\ u_H^{DG})_{L_2(\Omega)} = (v,\ u_H^{DG})_{L^2(\Omega)} \quad \forall u_H^{DG} \in V_H^{DG}$$

E_H. $\quad V_H^{DG} \to V_H := \{ v \in H_0^1(\Omega),\ v|_T \in P_1(T) \}$



$\Omega = [0,1]^2$

$H_0^1(\Omega)$

$$(E_H v)(z) = \begin{cases} 0 & z\text{是边界点} \\ \sum_{T \in T_H} \frac{v|_T(z)}{|N(z)|} & \text{内部点} \end{cases}$$

$$\frac{v|_{T_1}(z) + v|_{T_2}(z) + v|_{T_3}(z) + v|_{T_4}(z) \cdots v|_{T_6}(z)}{6}$$

6

定义拟插值算子 $I_H = E_H \circ \Pi_H^{dg}$.

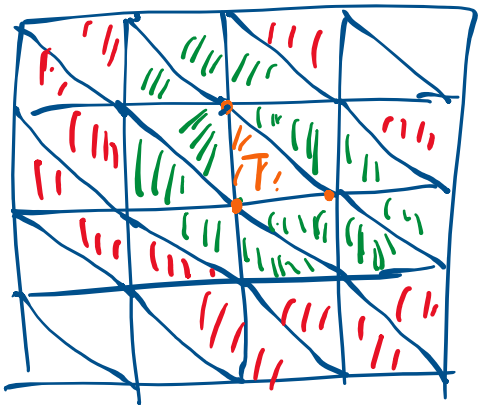
Thm 1. I_H 有如下性质 $I_H: H_0^1(\Omega) \to V_H$

① $I_H v_H = v_H$, $\forall\ v_H \in V_H$

② $\|v - I_H v\|_{0,T} \le C_I H \|\nabla v\|_{0,N_1(T)}$

$N_1(T) = \{T' \in T_H,\ T' \cap T \neq \phi\}$.

③ $\|\nabla I_H v\|_{0,T} \le C_I \|\nabla v\|_{0,N_1(T)}$



$N_1(T)$

$N_2(T) = 红 + N_1(T)$

定义 $W = \{ w \in V := H_0^1(\Omega),\ I_H w = 0 \}$.

$= \ker(I_H)$

二维 bilinear form

$a(u, v) = \int_\Omega (A\nabla u)\nabla v \, dx$

定义 V_H^{ms} 空间为 $a(\cdot,\cdot)$ 内积下 W空间的正交补

$$\boxed{V = V_H^{ms} \oplus W}$$

$$a(V_H^{ms}, w) = 0$$

定义算子 $C: V \to W$

$$a(Cv, w) = a(v, w) \quad \forall w \in W$$

下面刻画 V_H^{ms}空间

Thm $$V_H^{ms} = (I - C) V_H$$

proof 作业

理想化的CoD方法

Find. $u_H^{ms} \in V_H^{ms}$ st

$$a(u_H^{ms}, v) = \int_\Omega f v \cdot dx \quad \forall v \in V_H^{ms}$$

误差分析

Thm (3.4)

$$\|u - u_H^{ms}\|_{0,\Omega} \le \frac{N_0^2 C_I^2}{\alpha} H^2 \|f\|_{0,\Omega}$$

$$\|A^{\frac{1}{2}} \nabla (u - u_H^{ms})\|_{0,\Omega} \le \frac{N_0 C_I}{\sqrt{\alpha}} H \|f\|_{0,\Omega}$$

$$\|A^{\frac{1}{2}}\nabla(u-u_H^{ms})\|_{0,\Omega} \le \frac{N_0 C_I}{\sqrt{\alpha}} H \|f\|_{0,\Omega}$$

Proof.

设 u 是 model problem 的解，则 $u \in V$ 满足

$$a(u,v) = \int_\Omega f v \quad . \quad \forall v \in H_0^1(\Omega) \ (V) \quad (\Delta)$$

由于 $V_H^{ms} \subset V$

有 Galerkin 正交性

$$a(\underbrace{u-u_H^{ms}}_{\in V}, v_H^{ms}) = 0 \qquad \forall v \in V_H^{ms}$$

且有 $u - u_H^{ms} \in W$

即 $I_H(u-u_H^{ms}) = 0$

$$\|u-u_H^{ms}\|_{0,\Omega} = \|(I-I_H)(\underbrace{u-u_H^{ms}}_{v})\|_{0,\Omega}$$

N_0 有界常数

$$\overset{\text{Thm1}}{\underset{②}{\le}} N_0 C_I H \|\nabla(u-u_H^{ms})\|_{0,\Omega}$$

$$\le \frac{N_0 C_I}{\sqrt{\alpha}} H \|A^{\frac{1}{2}}\nabla(u-u_H^{ms})\|_{0,\Omega}$$

下面证

$$\|A^{\frac{1}{2}}\nabla(u-u_H^{ms})\|_{0,\Omega}^2 = a(u-u_H^{ms}, u-u_H^{ms})$$

$$\overset{\text{Galerkin}}{\underset{\text{正交性}}{=}} a(u-u_H^{ms}, u)$$

$$\overset{\text{正交性}}{=} a(u-u_H, u)$$

$$\overset{\text{对称性}}{=} a(u, u-u_H^{ms})$$

$$\overset{(\Delta)}{=} \int_\Omega f(u-u_H^{ms})$$

$$\leq \|f\|_{0,\Omega} \underline{\underline{\|u-u_H^{ms}\|_{0,\Omega}}}$$

即 · — — — —

(补充)

$$V_H^{ms} = \text{span}\{\Lambda_j - C\Lambda_j\}$$

Λ_j 是 V_H 的基函数

与一维不同的是 $C\Lambda_j$ 不再具有紧支集 且无法写出解析解

但从图 Fig 4.1 和 Thm 4.1 可看出

$C\Lambda_j$ 包含了问题的信息 且具有指数衰减性质

$\Downarrow$

构造 localized Correctors

定义 C_T $V \to W$

定义 $C_T \quad V \to W$

$$a(C_T v, w) = \int_T (A\nabla v)\nabla w \qquad \forall w \in V$$

则 $Cv = \sum_{T \in \mathcal{T}_H} C_T v$

但 $C_T v$ 仍然没有紧支集.

定义 $C_T^\ell \quad V \to W^\ell(T)$

$$a(C_T^\ell v, w) = \int_T (A\nabla v)\nabla w \qquad \forall w \in W^\ell(T)$$

$$W^\ell(T) = \{ w \in V,\ I_H w = 0,\ \mathrm{supp}(w) \subseteq N^\ell(T) \}$$

$N^\ell(T)$ 定义见 chap4 第1页 (P35)

定义 $C^\ell = \sum_{T \in \mathcal{T}_H} C_T^\ell$

$$C = \sum_{T \in \mathcal{T}_H} C_T$$

Thm 4.3

C^ℓ 与 C 之间具有指数衰减的逼近性质

$$V_{H,\ell}^{ms} = (I - C^\ell) V_H.$$

$$V_{H,\ell} = (I - C^{\ell}) V_H.$$

然而 显然 $W^{\ell}(T)$ 的函数具有局部紧支集

但 $W^{\ell}(T)$ 是无穷维的

二维时 $C_T^{\ell} \Lambda_j$ 没法写出解析解

因此 $V_{H,\ell}^{ms}$ 不能用作备散空间

令 $C_{T,h}^{\ell} : V_h \longrightarrow \underline{W^{\ell}(T) \cap V_h}.$

$$a(C_{T,h}^{\ell} v, w) = \int_T (A\nabla v)(\nabla w) \qquad \forall\, w \in W^{\ell}(T) \cap V_h$$

记 $C_h^{\ell} := \sum_{T \in T_H} C_{T,h}^{\ell}.$

$$V_{H,\ell}^{ms,h} = (I - C_h^{\ell}) V_H$$

Fully discrete LOD method.

Find $u_{H,\ell}^{ms,h} \in V_{H,\ell}^{ms,h}$ st

$$a(u_{H,\ell}^{ms,h}, v_{H,\ell}^{ms,h}) = \int_\Omega f v \qquad \forall\, v \in V_{H,\ell}^{ms,h}.$$

误差估计 Thm 5.5

误差估计　　Thm 5.5

程序实现　　chap7

basicLOD2d.m

数值算例

随机系数　　$f=1$

补充：　$h=2^{-4}, 2^{-8}$　FEM的解

$l=1,2$　　$H=2^{-3}, 2^{-4}$　LOD-FEM

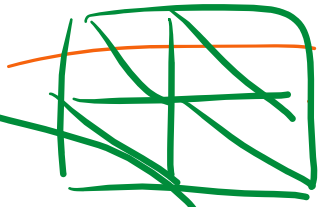
$H=2^{-2}$

A　$H=2^{-3}$

$H=2^{-4}$

$H=2^{-5}$

$l=1,2$ —



误差曲线

loglog (dof, L2Error)

(dof, HError)